

NorthPeak Data Platform - Production Runbook

Version: 1.0.0

Owner: Data Platform Engineering Team

Last updated: 14 July 2026

Emergency contact: Aisha Okafor (Data Engineering Manager)

1. Daily Operations Checklist

Start of day (09:00 UTC):

- ☐ Check Airflow DAG status at <http://airflow.northpeak.internal:8080>
 - ☐ Verify fact_sales row count for yesterday is greater than 100
 - ☐ Check Great Expectations checkpoint report for any failing expectations
 - ☐ Review quarantine table row count (should be less than 1% of total)
 - ☐ Confirm Power BI dashboard shows data for yesterday
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2. Common Incident Scenarios

Scenario A: FactSales missing data for a date

Symptoms: Power BI shows blank for a date. Chloe Tan will message you.

Investigation steps:

1. Check Airflow DAG run for that date: was it SUCCESS or FAILED?
2. If SUCCESS: check task-level logs for dbt_test_gold_models (known issue - see INC-001)
3. If FAILED: check which task failed and read the log
4. Check Bronze partition exists: `ls /bronze/np_core/orders/_bronze_ingestion_date=YYYY-MM-DD/`
5. Check Silver: `SELECT COUNT(*) FROM silver.orders WHERE DATE(order_date) = 'YYYY-MM-DD'`
6. Check Gold: `SELECT COUNT(*) FROM marts.fact_sales WHERE DATE(order_date) = 'YYYY-MM-DD'`

Fix - backfill:

```
airflow dags backfill northpeak_daily_sales \ -s 2026-10-12 \ -e 2026-10-14 \ --reset-dagruns
```

SLA: Resolve within 4 hours of detection. Escalate to Aisha if not resolved in 2 hours.

Scenario B: Bronze row count anomaly (too low)

Symptoms: Bronze row count alert fires. Less than 100 orders ingested for the date.

Investigation steps:

1. Check NP Core PostgreSQL connectivity: `psql -h np-core-prod.northpeak.internal -U pipeline_reader -d np_core -c "SELECT COUNT(*) FROM np_core.orders WHERE DATE(order_date) = CURRENT_DATE;"`
2. Check SupplyLink SQL Server connectivity via pyodbc
3. Check RewardsCo API health: `curl -H "Authorization: Bearer $REWARDSCO_API_KEY" https://api.rewardsco.com/v2/health`
4. Review weeks 38-40 anomaly pattern documented in process notes (known historical dip)

Do NOT: Assume low volume is a data quality error without checking source system first.

Scenario C: dbt test failure

Symptoms: Airflow dbt_test_gold_models task fails. Email alert fires.

Investigation steps:

1. Read the dbt test output: `cat /opt/northpeak/dbt/target/run_results.json`
2. Identify which test failed and which model it applies to
3. Common failures and fixes:

| Failing test | Likely cause | Fix |

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| assert_fact_sales_no_null_store_key | New store in NP Core not in DimStore | Run `dbt run --models dim_store` first |

| expect unique order_id | Dedup failed in Silver | Check `silver_cleansing.py` dedup logic |

| freshness error on silver.orders | Silver pipeline failed or ran late | Rerun `silver_cleanse_orders` task |

3. Backfill Procedure

Use this when: Missing data detected for one or more historical dates.

```
# Step 1: Identify missing dates SELECT date_day FROM marts.dim_date WHERE date_day >=
'2026-10-01' AND date_day NOT IN (SELECT DISTINCT DATE(order_date) FROM marts.fact_sales) ORDER
BY date_day; # Step 2: Run backfill for missing range airflow dags backfill
northpeak_daily_sales \ -s YYYY-MM-DD \ -e YYYY-MM-DD \ --reset-dagruns # Step 3: Validate
after backfill SELECT DATE(order_date), COUNT(*) as order_count FROM marts.fact_sales WHERE
DATE(order_date) BETWEEN 'YYYY-MM-DD' AND 'YYYY-MM-DD' GROUP BY DATE(order_date) ORDER BY
DATE(order_date);
```

4. Key Contacts

| Role | Name | Contact | When to contact |

|-----|-----|-----|

| Data Engineering Manager | Aisha Okafor | aisha.okafor@northpeak.co.uk | P1 incidents, all architecture decisions |

| Head of Data Platform | Marcus Webb | marcus.webb@northpeak.co.uk | Escalation if Aisha unavailable |

| BI Developer (dashboard) | Chloe Tan | chloe.tan@northpeak.co.uk | Dashboard issues, Power BI questions |

| Analytics Engineer | Yemi Adeyemi | yemi.adeyemi@northpeak.co.uk | dbt model questions, Gold layer issues |
| SupplyLink admin | Barry Higgins (IT) | barry.higgins@northpeak.co.uk | SQL Server connection issues only |

5. Credential Locations

NEVER commit credentials to GitHub.

| Credential | Location |

|-----|-----|

| NP Core DB password | Azure Key Vault: northpeak-kv / secret: np-core-db-pass |

| RewardsCo API key | Azure Key Vault: northpeak-kv / secret: rewardsco-api-key |

| NP Financial read replica cert | IT ticket NP-IT-2341, stored in Key Vault |

| Airflow admin password | LastPass: "NorthPeak Airflow Admin" |

6. Deployment Procedure (new pipeline changes)

```
# Step 1: Branch from main git checkout -b feature/NP-DATA-XXX-description # Step 2: Make changes, write tests pytest quality/pytest/ -v # Step 3: Push and open PR git push origin feature/NP-DATA-XXX-description # GitHub Actions runs pytest automatically on PR # Step 4: PR must pass all checks before merge # - pytest: all tests green # - dbt compile: no syntax errors # - Great Expectations: suite validates # Step 5: Merge to main triggers deployment to dev # Step 6: Manual promotion to prod after QA sign-off (Aisha)
```